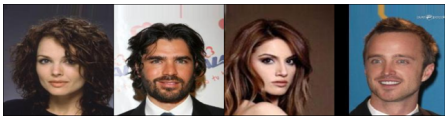
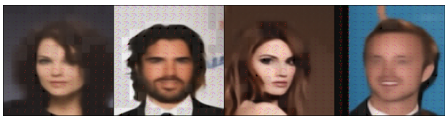
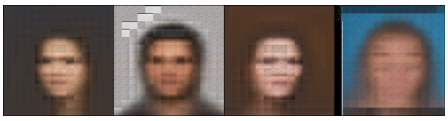


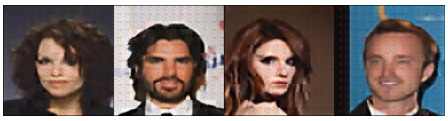
(a) Reconstructed from unprotected features.



(b) Reconstructed from features protected by GN [33].



(c) Reconstructed from features protected by GN+ (ours).



(d) Reconstructed from features protected by LP [15].

(e) Reconstructed from features protected by LP+ (ours).

Figure 3. Reconstruction results of model inversion attacks to features. '+' means the privacy-preserving technique is enhanced by our row permutation.

while the client holds private data, label, the embedding layer F_1, and the task head F_2. F_1 is computed locally on the client as the input data should be kept private from the untrusted server. F_2 is also with the client due to the labels are unknown to the server. The client sends the embeddings to the server and retrieves outputs of T for downstream tasks. The semi-honest server honestly follows the protocol but tries to reconstruct the private data from the embedding by model inversion attack.

We assess the utility of downstream tasks by classification accuracy and how private the client's embeddings are in protecting the inputs. The privacy is expressed by reconstruction metrics including Structural Similarity (SSIM), Peak Signal to Noise Ratio (PSNR) [13], and F-SIM of an MAE [11] inversion model [7, 10, 24]. The worse the reconstruction, the better the privacy protection. For more details of the setup, please see Appendix 11, which aligns with the threat model in [15, 33].

We perform row permutation to enhance the protection level of Gaussian Noise (GN), loss-pass filter (LP) with a radius of 0.05, and Batch Shuffle (BS) techniques. All baselines follow their original implementation. Two row permutation matrices of shape 197×197 are applied. The results are presented in Tab. 6. As shown, for GN and LP, the row permutation significantly enhances privacy while

Table 6. Utility and privacy on CelebA. ↓ means desirable direction. '+' means an enhanced version by our row permutation. Our methods are marked in light gray.

	Utility	Privacy		
	Acc /% ↑	SSIM ↓	PSNR ↓	F-SIM ↓
SL	91.9	0.645	16.25	0.933
GN [15]	88.8	0.457	15.47	0.663
GN+	88.8	0.167	8.068	0.187
LP [15]	90.1	0.450	14.71	0.614
LP+	90.2	0.110	5.687	0.154
BS [33]	90.8	0.148	8.180	0.176
BS+	91.1	0.143	7.613	0.167

maintaining the same level of utility as the baselines. For BS, both the utility and privacy are enhanced by our permutation way. The reconstruction visualization effect under model inversion attack is provided in Fig. 3 which clearly shows considerable privacy improvement.

To see how permutation affects the trade-off between accuracy and privacy, we report the trade-off curve of LP in Fig. 7 by varying the radius r from 0.01 to 0.05. Typically, with the decrease of the radius, the privacy is enhanced with the cost of utility: as the radius is reduced to 0.01, the SSIM, PSNR, and F-SIM scores decrease to 0.190, 7.950, and 0.196, respectively, while the accuracy drops to 85.6%. With shuffling, one does not need to reduce r to reach a high privacy level. For example, $r = 0.05$ achieves an accuracy of 90.2% under shuffling while reaching privacy no worse than $r = 0.01$ without shuffling. Overall, with row permutation, LP is able to obtain a better trade-off curve for all ranges of radiuses, as shown in Fig. 7, with high privacy levels at all times.

Model Encryption. Model parameter leakage, especially for proprietary large language models, could harm the intellectual property of the model owner. We consider an application that 'encrypts' the trained model weights by a permutation key, i.e., only the party who knows the permutation matrix can correctly use the model. As previously reported in Sec. 5.2, inference on the 'encrypted' model without the column-permutation key is close to random guesses. Further, we show that without the key, fine-tuning the 'encrypted' model is also ineffective.

We compare the training curves of three fine-tuning approaches. As a benchmark, the 'normal' approach fine-tunes the pre-trained models with normal procedures. Second, the 'authorized' approach refers to that the pre-trained model is permuted by Eq. 6 ($\boldsymbol{P}_C$ of shape 768×768) to obtain $\mathrm{T}_{(C)}$ before being fine-tuned with permutation (following Eq. 5). The permutation matrix $\boldsymbol{P}_C$ acts as an authorization key in the training process. Third, assume $\mathrm{T}_{(C)}$ is stolen by an unauthorized party who does not own $\boldsymbol{P}_C$. The unauthorized party then tries to fine-tune $\mathrm{T}_{(C)}$ with normal

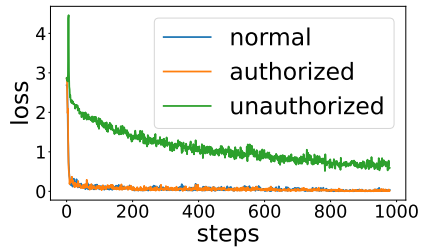


Figure 4. Training curves of fine-tuning ViT. The authorized has a performance close to normal while the unauthorized has a high loss.

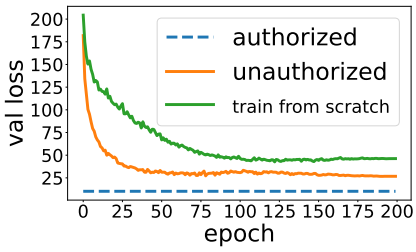


Figure 5. Validation loss curves of ViT trained to convergence. The unauthorized is far worse than the authorized but better than train-from-scratch.

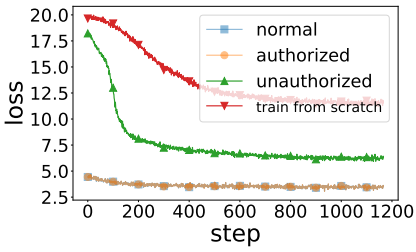


Figure 6. Training curves of fine-tuning GPT2. The unauthorized is far worse than the authorized but better than train-from-scratch.

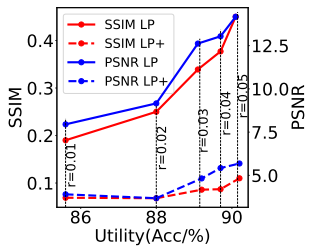


Figure 7. The utility-privacy trade-off curve by varying the radius of LP. The radius is denoted by r in the figure.

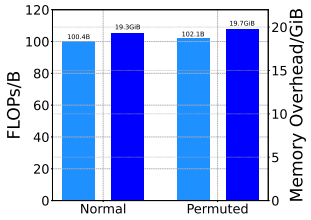


Figure 8. Comparison of computational and memory overhead.

procedures.

The testing accuracy of the fine-tuned models is displayed in Tab. 7 with their training curves until 1000 steps are provided in Fig. 4 for ViT-Base. The figure shows that the unauthorized can hardly learn as much as the authorized. To see if the unauthorized would further improve with a better training strategy, we train both the authorized and unauthorized until full convergence (of the authorized) in Fig. 5. The train-from-scratch model is trained with the same strategy, suggesting the lower bound of the unauthorized: in the worst case, the unauthorized party has to train random weights from the start, indicating no use of the 'encrypted' model.

It is observed that the performance of the unauthorized sits in between the authorized and that of training from scratch. A similar trend is observed in fine-tuning GPT2 on WikiText2 for text generation. The convergence curves are given in Fig. 6 where the authorized behaves almost identically to the normal. The unauthorized has a curve in between the normal and the train-from-scratch. Hence one can conclude that permutation indeed prevents anyone who is unaware of the permutation matrices from taking advantage of a trained model for inference or from releasing the full power of the pre-trained model by fine-tuning.

Table 7. Test accuracy (%) of the normal models fine-tuned in normal procedure, 'encrypted' models fine-tuned in authorized and unauthorized manners.

Model-FineTuning	ViT	Bert	GPT2
Normal-Normal	97.74	94.00	94.03
Encrypted-Authorized	97.94	93.72	93.66
Encrypted-Unauthorized	78.03	77.25	86.16

5.4. Efficiency

To see how much additional overhead our method incurs to vanilla learning, we evaluate the efficiency of our method by floating point operations (FLOPs) of the Transformer-Base backbone, and the memory consumption of training ViT-Base on Cifar10 in the normal and the permuted settings, respectively. For a more precise comparison, exactly four more additional matrix multiplications are applied to the permuted setting: two row permutation by $\boldsymbol{P}_R \in \mathbb{R}^{197\times197}$ and two column permutation by $\boldsymbol{P}_C \in \mathbb{R}^{768\times768}$. $\boldsymbol{P}_R$ is sampled per batch whereas $\boldsymbol{P}_C$ is fixed for the model. Results of Fig. 8 show that our method are almost as efficient as normal learning, incurring negligible overhead in computation and memory consumption.

6. Conclusion

We revise the concept of permutation equivariance and prove it as a property for Transformer-based models. The permutation equivariance of prior works is only a subset of ours: we show inter- and intra- token permutation equivariance in both forward and backward propagation. We not only theoretically analyze the property but also validate it through experiments. Finally, we propose two applications of this property: privacy-enhancing split learning and model authorization. Our research contributes to a deeper understanding of Transformer-based models and their wide potential applications.

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